Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/09/2016
PI Name:	Foster, Jamie Ph.D.		
Project Title:	Impact of Microgravity on the Cell-Cell Interactions between a Mutualistic Bacterium and Its Animal Host		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Merritt Island	**State:**	FL
Zip Code:	32953	**Congressional District:**	3
Comments:	EDITOR'S NOTE: Congressional district is for University of Florida, Gainesville. Address is PI's physical address.		
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	12/01/2013	**End Date:**	11/30/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2017 per NSSC information (Ed., 9/6/16)		
Key Personnel Changes/Previous PI:	June 2015: A new Ph.D. level graduate student joined the project. Sept 2014: As this was the first year of the award a postdoctoral fellow was hired.		
COI Name (Institution):			
Grant/Contract No.:	NNX13AM44G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Beneficial microbes play a significant role in the normal development of animal tissues. Through the formation of healthy stable associations between host and symbiont, particularly during post-embryonic development, carefully regulated developmental pathways can be induced. In this experimental research plan, we propose to examine the impact of microgravity on the normal molecular interactions between host and symbiont. To examine these effects we propose to use the model system between the squid Euprymna scolopes and the luminescent bacterium Vibrio fischeri. For over two decades this squid-vibrio model has served as an ideal ecosystem to examine the influences of mutualistic bacteria on animal development. To simulate microgravity, rotating bioreactors called high-aspect-ratio rotating wall vessel bioreactors (HARVs) will be used to provide a low-shear environment for both the host and symbiont, thereby simulating microgravity. Specifically, we propose the following: Objective 1. Examine the mechanism(s) associated with the delay of bacteria-induced immune cell trafficking that occurs in simulated microgravity. Objective 2. Characterize the changes in the transcriptome of the host squid in the presence and absence of symbiotic bacteria exposure under microgravity conditions. Objective 3. Develop a flight-ready microtiter plate for future flight experiments and optimize the conditions needed to initiate and maintain the symbiosis in this new hardware. The proposed work has several implications for the field of space biology. First, by examining how normal healthy symbioses are initiated and maintained under microgravity conditions, we can begin to determine the sensitivity of these ecosystems to perturbations in the space environment. Secondly, we can assess the genetic and metabolic responses to gravity in the presence and absence of symbiosis, potentially identifying previously unknown triggers or cues associated with post-embryonic animal development.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	All animals form symbiotic associations with bacteria. These ubiquitous interactions are essential for the normal development and health of the host organism. This project helps to improve our understanding of the molecular cross-talk that occurs between animals and microbes by delineating those genes and pathways that are influenced by gravity. As the force of gravity has not changed throughout Earth's history we have a unique opportunity to assess how these host-microbe interactions alter gene expression and their physiology in the absence of gravity.
Task Progress:	Task 1. Characterize the changes in the transcriptome of the host squid in the presence and absence of symbiotic bacteria under microgravity conditions. RNAseq - Generating transcriptomic libraries. We exposed host animals, in the presence and absence of symbiotic bacteria, to modeled microgravity conditions and generated transcriptomic libraries to each treatment (with gravity controls). Total RNA was extracted from the light organs (i.e., the site of symbiosis) and cDNA libraries were generated in triplicate. A total of 24 cDNA libraries were sequenced using the NextSeq500 generating over 100 GB of sequencing data. The libraries were generated and sequenced in YR2; however, the analysis of the data was completed in YR3. In YR 2 an analysis pipeline was developed to annotated and compare the sequenced derived from the RNAseq analysis as described in the previous report. In YR3 we have completed the analyses of the data and have identified the following effects associated with microgravity and host-microbe interactions: 1. Microbes appear to regulate host immune response in squid-vibrio association in modeled microgravity. Although the RNAseq analysis revealed several hundred significantly differentiated genes associated with modeled microgravity, there was a pronounced increase in the number of innate immune system related genes associated with aposymbiotic animals (i.e., animals that did not have their symbiotic bacteria) exposed to microgravity. These results suggest that microbes might be playing an important regulating the immune system of their host organisms during microgravity conditions. In addition to genes associated with the innate immune system there were also differentially regulated genes associated with oxidative stress (e.g., SOD1, superoxide dismutase), light organ remodeling and development (e.g., PCDHGB 6, 2, protocadherins; chymotryposins). A manuscript describing a full overview of the differentially expressed genes in the presence of microgravity is being submitted to Nature Microgravity in the next month. 2. Transcriptome assembly of host Euprymna scolopes – In addition to the microgravity studies, the results of this project are being used to generate an assembled transcriptome of the host organism Euprymna scolopes. The genome of E. scolopes is not currently available and the assembled transcriptome will help all researchers working on this model organism to improve their annotation and understanding of the host organisms. We are in the process of generating an online portal for researchers to access the data and compare the results to the E. scolopes database. In YR3 we used a state-of-the-art sequencing platform called PacBio Isoseq to expand our transcriptome data base and improve the quality and coverage of the host E. scolopes transcriptome. We generated 83,000 unique isoforms (i.e., genes) and are working in collaboration with the E. scolopes community genome project to help them improve the assembly of the squid genome. Task 2. Examine the mechanism(s) associated with the delay of bacteria-induced immune cell trafficking that occurs in simulated microgravity. A. Analysis of symbiont transcriptome in the presence and absence of the RNA-binding protein Hfq: Cultures of wild-type Vibrio fischeri and a mutant strain of V. fischeri defective in the gene hfq were exposed to simulated microgravity conditions using a High-aspect ratio bioreactor chambers. The gene hfq was targeted as previous research on known pathogens have identified this gene as a global regulator that is significantly down-regulated during simulated microgravity and actual space flight conditions. In YR2, total RNA was isolated from each treatment and then processed to remove ribosomal RNA. We then generated cDNA libraries to the remaining message RNA and then sequenced the libraries using a NextSeq Illumina platform. There were eight individual treatments and libraries were generated in triplicate for the treatments for a total of 24 cDNA libraries. In YR3 we have identified the expression of several Microbe-Associated Molecular Pattern (MAMPs) molecules such as lipopolysaccharide synthesis genes that appear to be upregulated in the absence of Hfq.

	B. Screening of V. fischeri mutant library to determine which genes are essential for microgravity and if they activate the host immune response. In addition to analyzing the V. fischeri transcriptome we have screened a transposon mutant library for any genes that may be essential for V. fischeri to live in microgravity conditions. Although not part of the original project, we had the opportunity to work with Dr. Mark Mandel of Northwestern University, who generated the library and we used the HARV reactor systems purchased through this Space Biology award. Essentially insertion sequencing was used to identify putative microgravity essential genes in V. fischeri. Transposon mutagenesis was performed in V. fischeri and this “input library” was then passaged under microgravity conditions twice. These libraries will be sequenced to identify which of the genes were critical for growing in a microgravity environment. Task 3. Delineate the flight hardware for future flight experiments and optimize the conditions needed to initiate and maintain the symbiosis during flight. Techshot Fluid Processing Cassette – Over the past three years I have met with numerous flight hardware companies about whether their hardware would be optimal for flying the squid-vibrio symbiosis. To conduct a flight experiment we need three compartments in which we can add animal hosts, the bacterial symbiont, and either RNAlater or a glutaraldehyde fixative. Based on these requirements I have narrowed down the best option for this system as the Techshot Fluid Processing Cassette (FPC). I have worked with Dave Reed of Techshot regarding the FPC and we have assessed the biocompatibility of the Fluorinated Ethylene Propylene (FEP) bags and the Pharmed BPT tubing as well as the population loading of the bags. Currently, the FPC system seems to be the most ideal hardware for replicating many of these ground experiments in high quality microgravity conditions. During our no cost extension we plan to continue these experiments to optimize the number of host animal replicates within the FEP bags as well as how long the animals can be maintained within the hardware.
Bibliography Type:	Description: (Last Updated: 04/20/2026)
Articles in Peer-reviewed Journals	Heath-Heckman, EA, Foster JS, Apicella MA, Goldman WE, McFall-Ngai MJ. "Environmental cues and symbiont MAMPs function in concert to drive the daily remodeling of the crypt-cell brush border of the Euprymna scolopes light organ." Cellular Microbiology. 2016 Nov;18(11):1642-1652. Epub 2016 May 3. https://doi.org/10.1111/cmi.12602 ; PubMed PMID: 27062511; PubMed Central PMCID: PMC5219934 , Nov-2016